A Fast Commutation Error Correction Method for Sensorless BLDC Motor Considering Rapidly Varying Rotor Speed

Hao Jin, Gang Liu, *Member, IEEE*, Haitao Li, Baodong Chen, and Haifeng Zhang

***Abstract*—Accurate commutation point is essential to ensure the ideal torque performance and low power consumption for brushless DC (BLDC) motor. In order to achieve the precise sensorless commutation at a constant or the rapidly varying speed, a fast commutation error correction method is proposed in this article. First, it is pointed out that the controlled variable variation, the high rate of commutation error relative to the low speed and the nonlinearity of the correction loop are the main origins of commutation error when the speed varies rapidly. Then, a novel commutation error correction method is proposed through the following three aspects: first, constructing a controlled variable first utilized for commutation correction, which keeps constant at the rapidly varying speed; second, incorporating feedforward compensation for the phase shift caused by low-pass filter; third, using feedback linearization to reduce the nonlinearity of the correction loop. The proposed commutation error correction method not only corrects the commutation error fast at a constant speed, but also achieves precise commutation when the rotor speed changes rapidly. Moreover, the proposed method is based on the principle of closed-loop control, thus, it also has the advantage of anti-interference. At last, the experimental results are shown to validate the effectiveness of the proposed method.**

***Index Terms*—Commutation error compensation method, feedforward control, sensorless drive.**

I. Introduction

OWNING to the advantages of high energy density and simple structure, the BLDC motors are widely used in industry [1], [2]. The digital Hall-effect sensor or photoelectric encoder are commonly employed as position sensors to implement the rotation, but the output signals can be affected by electromagnetic interference and mounting accuracy [3], [4]. Therefore, numerous sensorless drive methods have been further researched in the last decades.

Among the sensorless techniques, the direct detection method is usually used due to the simple configuration. The main idea is based on the position relationship between the motor signals and commutation points. The back-electromotive force (back-EMF) detection strategy is typical of the direct detection method, and the commutation point is determined from zero-crossing-points (ZCPs) of back-EMFs delayed by 30° [5]. Similarly, the line-to-line back-EMF is another common signal to detect, the ZCPs of which are right at the points for commutation [6], [7]. In addition, there are the G function detection method [8], [9], virtual neutral voltage detection method [10], etc. However, the LPFs are inevitably adopted to remove the noise from PWM waveform and freewheel current. Thus, the phase shift as the commutation error is introduced. Even though it can be approximately compensated from measurement or calculation, it is an open-loop commutation approach without anti-interference ability. Based on the direct detection method, a closed-loop controller can realize the real-time correction of commutation positions. Such kind of closed-loop correction method has become a research hot-spot in recent years.

The research of the conventional closed-loop correction method is mainly carried out from the following two catalogues: the construction of controlled variable and the optimization of control algorithm. The construction of controlled variables is based on the motor signals symmetry and periodicity. In [11] and [12], before and after the active duration of one phase, the phase voltage difference is employed as the controlled variable. Similarly, in [13], before and after the 60° interval when a certain phase is inactive, the floating back-EMF is sampled for correction. Furthermore, the integral motor signals can also be adopted to implement the closed-loop correction, such as the floating phase back-EMFs integral [14], virtual neutral point voltage integral [15]. Besides the voltage signal, the phase current can also be employed to correct the commutation error [16]–[20].

Due to the nonlinear property of the correction loop, the optimization of the control algorithm has also been further studied. The model free adaptive control algorithm is used instead of the PI controller and improves convergence speed [21]. The

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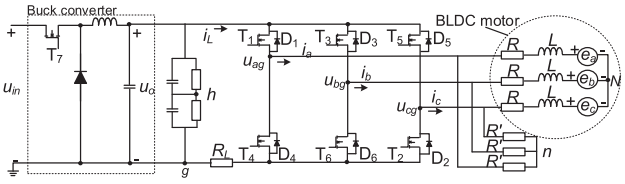


Fig. 1. Topology of the inverter with buck converter.

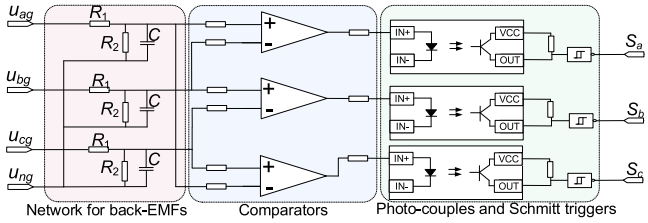


Fig. 2. Line-to-line back-EMFs ZCPs detection circuit. R_1, R_2, and C constitute the LPFs.

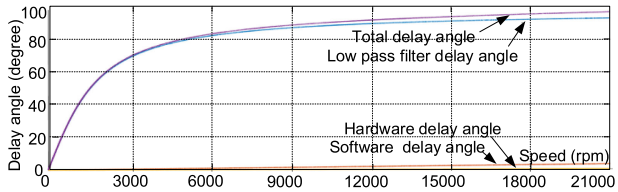


Fig. 3. Trend of different commutation errors relative to rotor speed.

work in [22] presents a correction method using fuzzy-neural-networks, and the robustness of correction is improved. An adaptive compensation method based on the radial basis function (RBF) neural network has also been studied, which improves the correction dynamic performance [23].

But the motor signals are distorted at the rapidly varying speed, which leads the conventional method to be unsuitable. However, to the best of authors' knowledge, there are seldom works focusing on this issue. In essence, the issue can be regarded as a closed-loop correction for commutation with the distorted motor signals. For a motor with three-phase asymmetric back-EMFs, Chen and Liu exploit a low-pass filter to extract the fundamental waves of the back-EMFs, and detect the intersection positions for commutation [24]. Similarly, a least mean square based Adaline filter is used to acquire the d-axis current, and the angle between the vectors of filtered current and back-EMF is calculated for correction [25]. In [26], the second-order generalized integrator based phase-locked loop is used to estimate the rotor position, and a synchronic-frequency filter is set as a prefilter. However, the filtered signals mentioned above can be distorted by varying speed as well. In order to overcome the drawbacks, a fast commutation error correction method is proposed in this work.

The main idea of the proposed method includes the following three contents: first, utilizing a controlled variable that can keep constant at varying speed, utilizing feedforward compensation, and feedback linearization to optimize correction algorithm. The proposed method not only presents fast convergence behavior at a constant speed, but also achieves high-precision commutation when the rotor speed changes rapidly.

II. Analysis of Commutation Error for Sensorless BLDC Motor

A. BLDC Motor Mathematical Model and Line-to-Line Back-EMF ZCPs Detection Method

In order to reduce the eddy current loss, an air-core stator with small inductance is adopted in BLDC motor in this work. For this reason, the buck converter (see Fig. 1) is employed to reduce the switching noise, and the mutual inductance between phases can be ignored. Thus, the mathematical model can be written as

$$u_{xg} = Ri_x + L\frac{di_x}{dt} + e_x + u_{Ng} \tag{1}$$

where the subscript x denotes a, b, or c corresponding to three phases, u_{xg} is the terminal voltage, e_x is the back-EMF, i_x is the phase currents, u_{Ng} is the voltage of the neutral point, R is the phase resistance, and L is the phase inductance.

The line-to-line back-EMF ZCPs detection method is adopted for sensorless commutation [20]. The detection circuit is shown in Fig. 2, which consists of LPF, comparator, photocoupler, Schmitt trigger, etc. The amplitude-frequency characteristics $(G_{\text{filter}}(\omega))$, phase-frequency characteristics $(\varphi_{\text{filter}}(\omega))$, and cut-off frequency (f_{filter}) of the LPF are described as

$$\begin{cases} G_{\text{filter}}(\omega) = \frac{1}{\sqrt{(R_1C\omega)^2+(1+R_1/R_2)}} \\ \varphi_{\text{filter}}(\omega) = -\arctan(\frac{R_1C\omega}{1+R_1/R_2}) \\ f_{\text{filter}} = \frac{R_1+R_2}{2\pi R_1R_2C} \end{cases}. \tag{2}$$

Thus, there is phase shift on virtual Hall signals from (2). Besides, the hardware and software can also introduce the commutation delayed angle. Therefore, the total commutation error can be described as

$$\varphi = -\varphi_{\text{filter}} + \varphi_h + \varphi_s \tag{3}$$

where φ_h and φ_s denote the delayed angle from hardware and software, respectively. φ_{filter} denotes the delayed angle from the LPF.

The hardware and software delay angles in terms of rotor speed can be approximately obtained by calculation according to electronic component datasheet and measurement of software running time. Thus, the approximate relationship between the commutation error and the motor speed is obtained as Fig. 3. We can see that the phase shift generated from the LPF accounts for the main proportion of total error, especially in the low-speed region. In order to achieve precise commutation, numerous closed-loop correction methods have been developed. However, the unsuitable controlled variable and control method will introduce phase error at rapidly varying speed.

B. Commutation Error Analysis for the Conventional Method at Rapidly Varying Speed

1) Commutation Error Generated by Changes of Controlled Variable: As mentioned in Section I, the motor signals have the properties of aperiodicity and asymmetry at the rapidly varying speed. As shown in Fig. 4, the unsuitability of the

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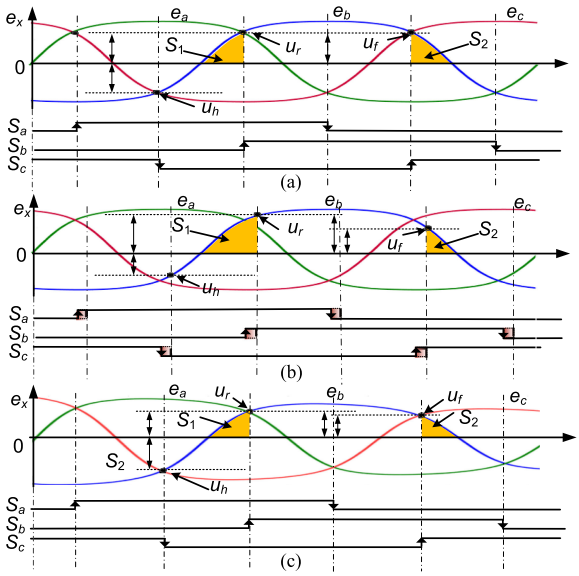


Fig. 4. Back-EMFs and virtual Hall signals with different commutation points. (a) Accurate commutation. (b) Delayed commutation at a constant speed. (c) Accurate commutation at decreasing speed.

conventional method in such scenario is explained. To make a comparison, the conventional methods proposed in [11]–[14] are analyzed as follows.

1) Fig. 4(a) shows the commutation points are at the back-EMFs intersections, which denotes that the commutation is accurate at a constant speed [27]. The floating back-EMF of phase B as the feedback signal is symmetrical and periodical. According to [11] and [12], the controlled variable for commutation correction is Δu_1($\Delta u_1 = u_f - u_r$ where u_r and u_f are voltages sampled from e_b at commutation points), and the equation $\Delta u_1 = 0$ can be obtained. In [12], $\Delta u_2 = u_r + u_h = 0$ is deduced (where u_h is the voltage sampled from e_b at the commutation point). Similarly, $\Delta S = S_2 - S_1 = 0$ is derived from [14] (where S_2 and S_1 are the integrals of floating phase back-EMFs).
2) Fig. 4(b) shows the commutation is delayed at a constant speed, and the symmetry and periodicity of the floating back-EMFs is distorted. The controlled variables satisfy the following expressions: $\Delta u_1 < 0$, $\Delta u_2 < 0$, and $\Delta S < 0$. With respect to the advanced commutation, $\Delta u_1 > 0$, $\Delta u_2 > 0$, and $\Delta S > 0$ can be deduced. After the conventional method introduced, the controlled variables converge to zero [see Fig. 4(a)]. Meanwhile, the virtual Hall signals are corrected.
3) Fig. 4(c) shows the commutation is accurate at decreasing speed, and the three-phase floating back-EMFs are asymmetrical and aperiodical. The equation of $\Delta u_1 < 0$, $\Delta u_2 < 0$, and $\Delta s < 0$ can be deduced. Thus, the advanced angle as the commutation error is generated improperly using the conventional method.

From the analysis mentioned above, the conventional method can present a good performance at constant speed. However, the controlled variables will change with the varying speed causing the improper correction angle.

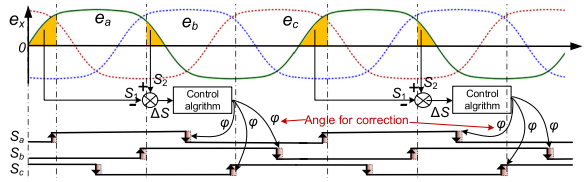


Fig. 5. Diagram describing that the conventional method is performed only once per electrical cycle.

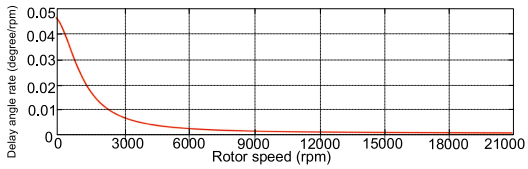


Fig. 6. Rate of LPF delay angle relative to the speed.

In addition, only one of three-phase back-EMFs is employed to correct the three virtual Hall signals according to the conventional method (see Fig. 5). Thus, the correction is performed only once in an electrical cycle. However, there are six commutation points during the interval. Hence, the frequency for correction is low, which results in a poor dynamic performance.

2) Commutation Error Analysis in Low-Speed Region: Fig. 6 shows the rate of LPF delay angle relative to the speed, and it is greater in the low-speed range. Considering that the delay angle from LPF accounts for the main proportion of total commutation error, the total commutation error decreases fast if the motor decelerates rapidly in the low-speed range. Owning to the poor dynamic performance of the conventional method, the variation of the commutation error cannot be corrected in time resulting in phase error on virtual Hall signals.

In summary, when the conventional method is employed at the rapidly varying speed, inaccurate commutation is attributed to the following reasons: 1) The varying speed leads to the change of the controlled variables, which results in improper correction. 2) The commutation error angle changes rapidly with the speed, which cannot be corrected in time for the poor dynamic performance.

III. Novel Fast Commutation Error Correction Method

Based on the previous analysis, the correction method can be optimized from the following three perspectives: avoiding the impact of speed variation on the controlled variables, preventing interference from fast changing commutation error in low-speed range, and enhancing the dynamic performance of the correction. For that purpose, the proposed method covers the following three approaches.

A. Construction of Controlled Variable First Utilized for Commutation Correction

A controlled variable that does not change with the varying speed is first proposed. From (1), u_{nh} can be derived as

$$u_{nh} = \frac{u_{ag} + u_{bg} + u_{cg}}{3} - \frac{u_o}{2} \tag{4}$$

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TABLE I
RELATIONSHIP BETWEEN u_{nh} AND COMMUTATION

Condition	Conducted/Switching phases	u_{nh}
Conduction	AB, BA	$\frac{1}{3}(e_c - \frac{e_a+e_b}{2})$
	BC, CB	$\frac{1}{3}(e_a - \frac{e_b+e_c}{2})$
	CA, AC	$\frac{1}{3}(e_b - \frac{e_a+e_c}{2})$
Commutation	$AB-AC, BC-BA, CA-CB$	$\frac{u_o}{6}$
	$CB-AB, AC-BC, BA-CA$	$-\frac{u_o}{6}$

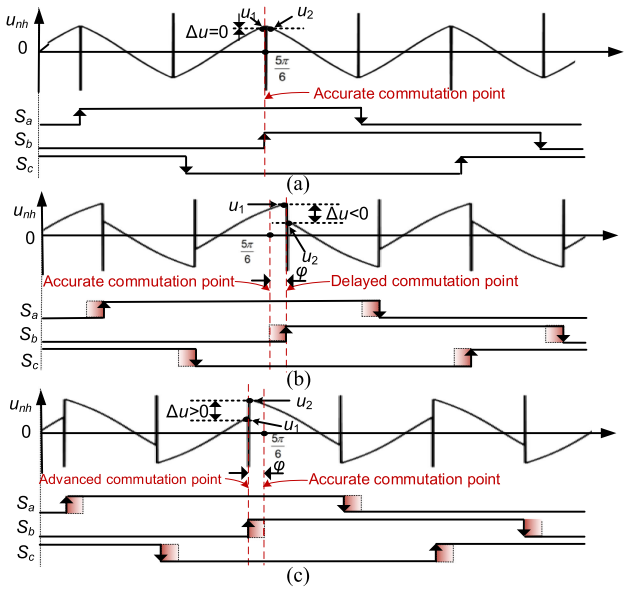


Fig. 7. Positional relationship between u_{nh} and S_x with different commutation points. (a) Accurate commutation. (b) Delayed commutation. (c) Advanced commutation.

where u_o is the dc-link voltage, and u_{nh} is the voltage from the dc-link voltage middle point to the virtual neutral point.

Assume that phases A and C are conducted. The three-phase terminal voltages can be described as

$$\begin{cases} u_{ag} = 0 \\ u_{bg} = e_b + u_{Ng} \\ u_{cg} = u_o \end{cases}. \tag{5}$$

Based on (5) and (1), the neutral point voltage can be expressed as

$$u_{Ng} = \frac{1}{2}(u_o - e_a - e_c). \tag{6}$$

Base on (4)–(6), u_{nh} can be written as

$$u_{nh} = \frac{1}{3}\left(e_b - \frac{e_a + e_c}{2}\right). \tag{7}$$

During the freewheeling interval after commutation, there are $u_{ag} = 0, u_{cg} = u_o$, and $u_{bg} = u_o$, thus, $u_{nh} = -u_o/6$. Therefore, u_{nh} can be represented by Table I. If the commutation freewheeling process is not considered, the voltage switching positions of u_{nh} are synchronized with commutation points from Table I. Fig. 7 shows the relationship between u_{nh} and S_b edges,

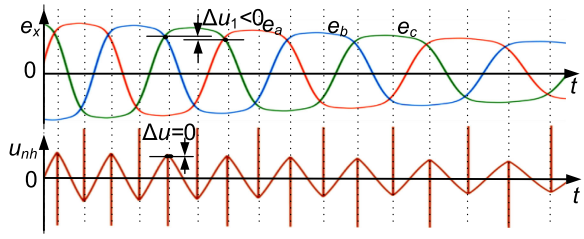


Fig. 8. Difference between the controlled variable acquired from three back-EMFs and that from u_{nh} at the decreasing speed.

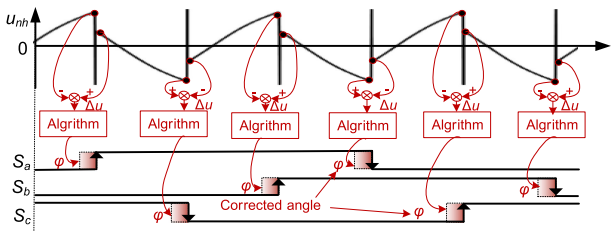


Fig. 9. Diagram illustrating that the proposed method with the new controlled variable can be performed for 6 times during an electrical cycle.

and u_{nh} is sampled on the left and right sides of the virtual Hall signal edges, the sampled voltages of which are defined as u_1 and u_2, respectively. The voltage difference is defined as

$$\Delta u = u_2 - u_1. \tag{8}$$

When the commutation is accurate [see Fig. 7(a)], delayed [see Fig. 7(b)] and advanced [see Fig. 7(c)], there are $\Delta u = 0$, $\Delta u < 0$, and $\Delta u > 0$, respectively. Moreover, there is a negative correlation between φ and Δu, thus, Δu is utilized as the controlled variable to implement the closed-loop correction in this work. When Δu converges to zero, the commutation error is corrected at the same time.

B. Feedback Linearization for the Correction Loop

Taking the commutation process from AC phase conduction to AB phase conduction as an example, the controlled variable can be expressed as

$$\Delta u = \frac{1}{3}\left(e_b - \frac{e_a + e_c}{2}\right) - \frac{1}{3}\left(e_c - \frac{e_a + e_b}{2}\right) = \frac{1}{2}(e_b - e_c). \tag{9}$$

The following two conclusions can be drawn from (9).

1) When the commutation is accurate, Δu remains zero even though the back-EMF waveforms are distorted. As shown in Fig. 8, the switching positions of u_{nh} are synchronized with back-EMFs intersections, and Δu is equal to zero when the motor decelerates. Therefore, the closed-loop correction fed by Δu can avoid the interference from varying speed. Besides, the correction process can be performed for 6 times (see Fig. 9) during one electrical cycle, which increases the correction frequency and presents a better dynamic performance. On the contrary, Δu_1 as the

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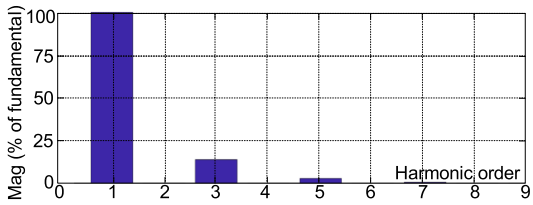


Fig. 10. Harmonics of back-EMFs waveform.

controlled variable mentioned in [12] and [13] achieves a negative value, thus, the advanced compensated angle is improperly generated.

2) The amplitude of controlled variable is half of back-EMF voltages deviation at the sampling points. Since the back-EMF consists of harmonics with different orders and amplitudes, the nonlinear relationship between φ and Δu is determined. This is another attribution for the poor dynamic performance [12], [22].

Commonly, the back-EMF waveforms of BLDC motor are between a sine wave and trapezoidal wave, which can be described by the following equation [25]:

$$\begin{cases} e_x = e_{x1} + e_{x3} + e_{x5} + \cdots \\ = \sum_{k=1}^{\infty} \cdot \sin((2k-1) * (\theta + m \cdot 2\pi/3)) \\ A_{2\text{-}k-1} = k_{2\text{-}k-1} \cdot \omega \\ \sum_{a,b,c} e_{x1} = \sum_{a,b,c} e_{x3} = \sum_{a,b,c} e_{x5} = \cdots = 0 \end{cases} \tag{10}$$

where $2k-1$ denotes the harmonic order, and m is -1, 0, or 1, which determines the different phases. k_{2k-1} is the harmonic constant, and A_{2k-1} is the coefficient for different harmonic. θ is the electrical angle.

Based on (10), the fast Fourier transform (FFT) of the back-EMF is shown in Fig. 10. It can be seen that the fundamental and third harmonic account for main proportion. Therefore, the harmonics within the third order can be utilized to acquire the approximate relationship between φ and Δu. Taking the commutation point θ ($\theta = \frac{5\pi}{6} + \varphi$, Fig. 7) as an example, the controlled variable is derived as

$$\Delta u \approx \frac{\sqrt{3}A_1}{2} \sin(\varphi) \tag{11}$$

where A_1 is the first harmonic coefficients. From (11), the approximate expression between φ and Δu can be deduced as

$$\varphi \approx \arcsin\left(\frac{2\Delta u}{\sqrt{3}A_1}\right). \tag{12}$$

With (12) incorporated into the feedback channel, φ can be employed as a new controlled variable, which transforms the property of the correction loop from nonlinearity to linearity. Then, the control performance can be easily optimized to improve the correction convergency speed and robustness. The PI regulator fed by φ is finally adopted as

$$\varphi_c = k_p(\varphi + k_i \int \varphi dt) \tag{13}$$

TABLE II
MOTOR PARAMETERS

Content	Unit	Value
Power supply	V	310
Rotating speed	rpm	12000
Stator resistance	Ω	0.43
Stator inductance	mH	0.35
Pole pairs	\	2

where k_p is the proportional coefficient, and k_i is the integral coefficient. φ_c is the angle for correction. With the output of the PI control algorithm, the virtual Hall signals can be corrected in real time.

C. Feedforward Compensation for the Phase Shift From LPF

As analyzed earlier, the commutation error angle changes fast with the rapidly varying speed in the low-speed range. Considering that the LPF delay angle accounts for the main part of the commutation error, the online calculation result of LPF phase shift can be used as the feedforward compensation angle. From (2), it can be written as

$$\varphi_f(\omega) = \arctan\left(\frac{R_1 C\omega}{1 + R_1/R_2}\right). \tag{14}$$

With φ_f added into the output of PI controller, the commutation error induced from speed variation can be feedforward compensated mostly.

Fig. 11 demonstrates the basic block diagram of the proposed method. The feedback linearization block transform Δu to φ, and φ_f is calculated as the feedforward compensation angle to avoid the impact from the LPF. Shifted by the correction angle ($\varphi_f + \varphi_c$), the virtual Hall signals obtained from line-to-line back-EMFs are corrected through the phase-shift block.

IV. EXPERIMENTAL VERIFICATION

A. Experimental Setup

As Fig. 12 shown, a platform of the magnetically suspended turbo molecular pump (TMP) using active magnetic bearings was constructed to verify the proposed method. The active magnetic bearings have the advantage of no contact, which allows the BLDC motor of TMP operates at high speed. As shown in Fig. 13, the back-EMF is in nonideal trapezoidal waveform. The parameters of the BLDC motor are listed in Table II. The digital controller is TMS320F28335 with an intelligent power module (PM50RL1B060). Considering that the controlled variable is too low to be detected for commutation error correction, the sensorless "three-step startup" method of BLDC motor is utilized for start-up in this work [13]. After the prepositioning stage, the rotor is dragged to a special angle. Then, the BLDC motor starts to rotate with the increasing phase voltage and commutation frequency in the external synchronization stage. Finally, the BLDC operates with the proposed method in the self-synchronization stage.

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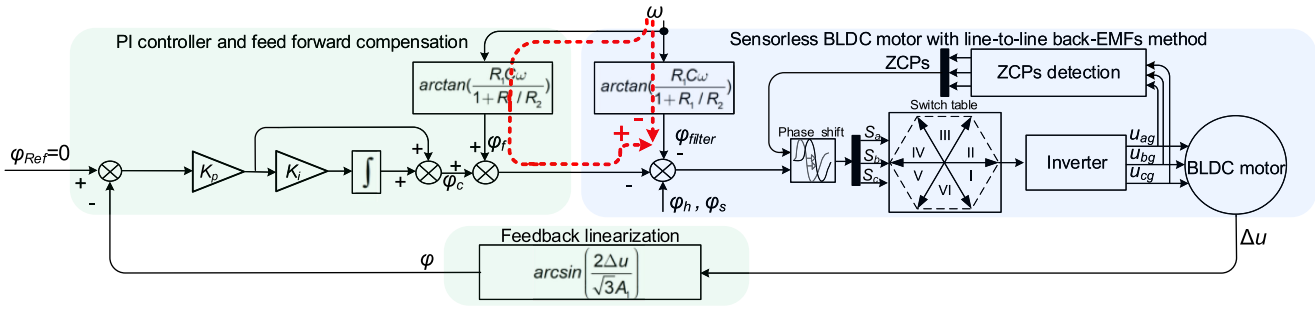


Fig. 11. Schematic diagram of the sensorless BLDC motor and the fast closed-loop correction method proposed in this work.

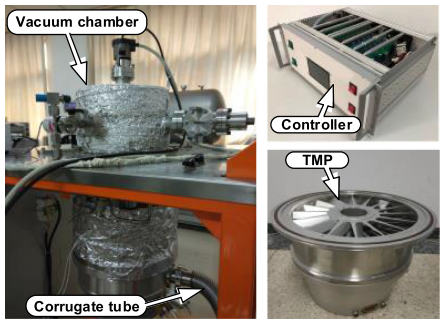


Fig. 12. Experimental platform of magnetically suspended TMP.

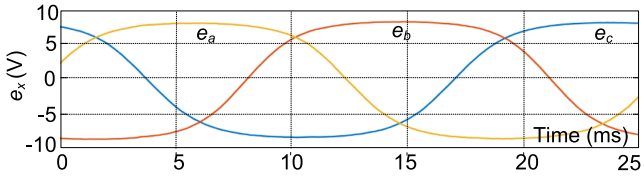


Fig. 13. Back-EMFs of the BLDC motor in this work.

B. Experiment 1: Verification for the Controlled Variable Suitability at Rapidly Varying Speed

In order to verify that the voltage switching points of u_{nh} are synchronized with the intersections of back-EMFs, and Δu remains to be zero at a constant or rapidly changing speed with the accurate commutation, the Hall sensors are employed to provides precise rotor position information. When the phase x is not conducted or the dc-link voltage is zero, the phase current satisfies $i_x = 0$. Substituting it into (1) gives

$$u_{xN} = u_{xg} - u_{Ng} = e_x \tag{15}$$

where u_{xN} is the phase voltage.

From (15), the crossing points of u_{aN}, u_{bN}, and u_{cN} are the back-EMFs intersections. Because of the ultrahigh vacuum environment kept by the backing pump in TMP, the motor speed can be treat as a constant with the dc-link voltage switched OFF in short time. Then, the motor decelerates immediately by air drag force with the air valve opened. Fig. 14(a) depicts the experiments based on the above approaches at 12 000 r/min. The variation of u_{nh} and u_{xN} includes three stages. First, the speed remains to be constant controlled by the dc-link voltage. Then, the dc-link voltage is switched OFF with the rotor speed keeps constant within a short time. Finally, the speed decreases for the increasing of load.

As shown in Fig. 14(a) and (b), the switching positions of u_{xN} are synchronized with the intersections of u_{xN} at the constant speed. The same conclusion can be drawn for the motor deceleration process [see Fig. 14(c)]. Furthermore, Δu is equal to zero during the three stages, which indicates that Δu is suitable for commutation correction at the constant and varying speed.

C. Experiment 2: Verification of the Proposed Method at a Constant Speed

Fig. 15 shows the waveforms of u_{nh}, S_a, H_a, and i_a at 12 000 r/min, where H_a is the Hall signal as reference for evaluating the proposed method performance. When the commutation point is delayed by φ [see Fig. 15(a)], a phase deviation between S_a and H_a is introduced, and the waveforms of i_a and u_{nh} are distorted. Thus, the sampled voltages of u_{nh} are different leading Δu to be negative. On the contrary, when the commutation points is advanced [see Fig. 15(b)], i_a and u_{nh} are also be distorted, and S_a leads H_a by φ. Δu achieves a positive value. As the proposed method is adopted [see Fig. 15(c)], the commutation error is eliminated with Δu converging to zero. In addition, Δi is significantly reduced. Thus, the higher harmonics of the phase current are weakened, which improves the motor efficiency.

Fig. 16 shows the fast FFT of dc-link current when the commutation is inaccurate and corrected. It can be found that if the commutation is advanced [see Fig. 16(a)] or delayed [see Fig. 16(b)], there are harmonics with higher amplitude in dc-link current. As shown in Fig. 16(c), the amplitude of harmonics decreases with the introduction of the proposed method, thus, the motor efficiency is improved.

D. Experiment 3: Dynamic Performance and Anti-Interference Ability Verification

1) Dynamic Performance Verification: The conventional method proposed in [12] and [23] is implemented to make a comparison. Fig. 17(a) and (b) shows the commutation error convergence process using the conventional method and the proposed method at 600 r/min, respectively.

It can be seen from Fig. 17(a) that the delay angle (approximate to 18°) is between S_a and H_a before t_1. Δu_1 is negative, and the distortion of i_a is obvious. After the conventional method

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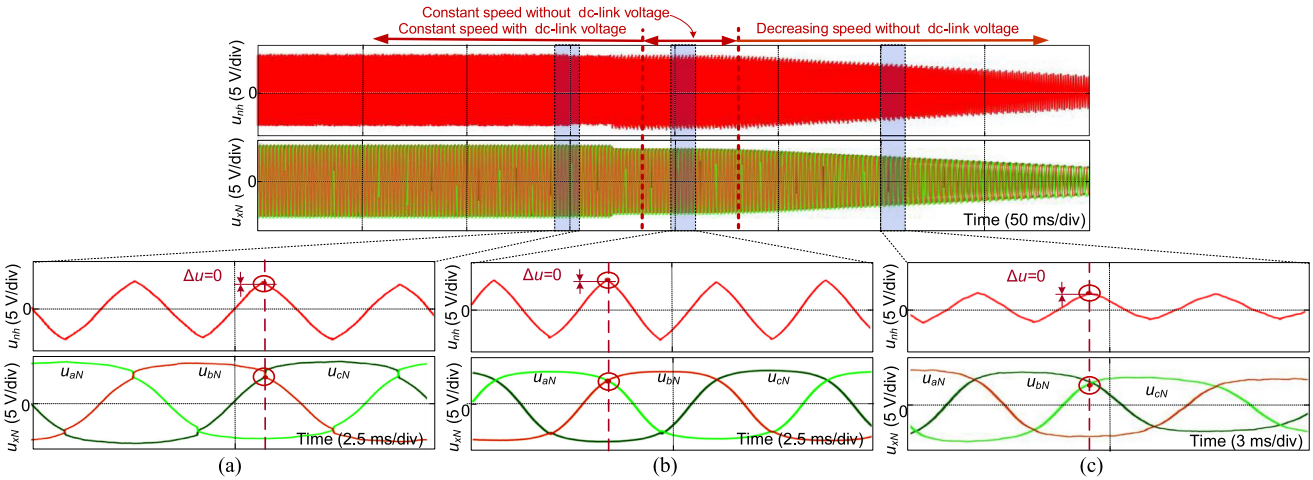


Fig. 14. Verification that the u_{xN} intersection points are synchronized with the positions where u_{nh} switches at constant speed or decreasing speed. (a) Constant speed with dc-link voltage. (b) Constant speed without dc-link voltage. (c) Decreasing speed without dc-link voltage.

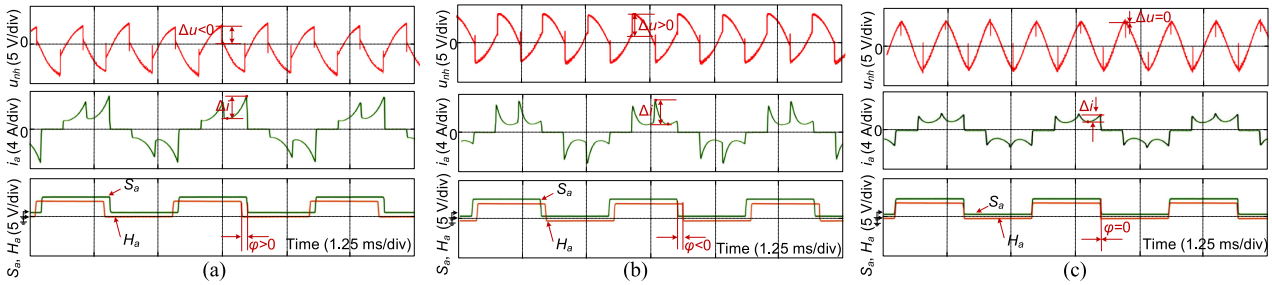


Fig. 15. Waveforms of u_{nh}, S_a, H_a, and i_a at the speed of 12 000 r/min with different commutation points. (a) Delayed commutation. (b) Advanced commutation. (c) Commutation with the method proposed in this article.

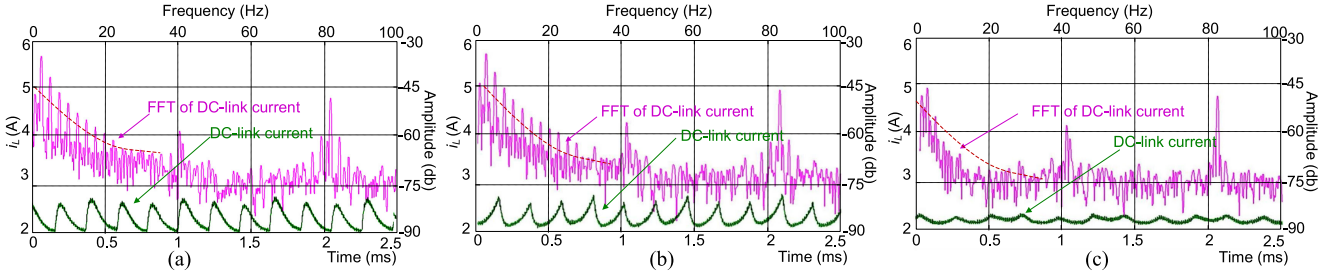


Fig. 16. FFT of dc-link current of motor with different commutation error at 1200 r/min. (a) Advanced commutation. (b) Delayed commutation. (c) Commutation with the proposed method.

employed at t_1, the phase deviation between S_a and H_a is eliminated with Δu_1 converging to zero. Similarly, as shown in Fig. 17(b), after the proposed method adopted at t_2, S_a is synchronized with H_a gradually, and Δu converges to zero with phase error corrected. However, the convergence speed of the two methods is different. It takes about 18 electrical cycles for the convergence of the conventional method in contrast to 4 electrical cycles with the proposed method. This indicates that the proposed method has the stronger dynamic performance and can achieve fast correction of commutation error.

2) Anti-Interference Ability Verification: In order to verify the proposed method anti-interference ability. As shown in Fig. 18, the virtual Hall signals are corrected before t_3. Then, the corrected angle is suddenly cleared in the controller. Consequently, i_a and u_{nh} are distorted. However, i_a and u_{nh} converge to the symmetric state gradually, and S_a synchronizes with H_a again after 4 electrical cycles. That means the commutation error is finally corrected for the proposed method anti-interference ability.

E. Experiment 4: Verification of Proposed Method at Rapidly Varying Speed

In order to verify that the proposed method can achieve high-precision commutation at rapidly varying speeds, the motor

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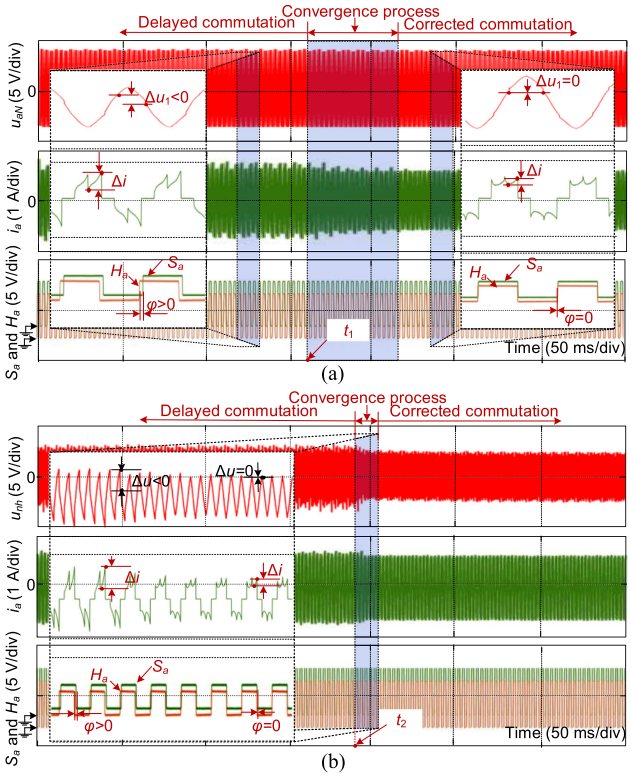


Fig. 17. Commutation error convergence process comparison between different correction method at 600 r/min. (a) Conventional method. (b) Proposed method.

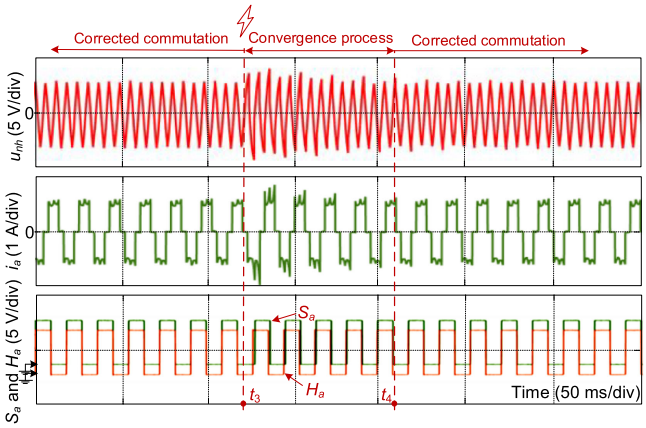


Fig. 18. Variation of u_{nh}, i_a, S_a, and H_a when the corrected angle is cleared at t_3.

speed was reduced by suddenly increasing load at 1200 r/min and 12 000 r/min, respectively.

The rate of the commutation delay angle respective to speed is high at 1200 r/min (see Fig. 6). Thus, if the speed decreases rapidly, the fast change of commutation error will act as the interference to the closed-loop correction. Fig. 19(a) shows the waveforms of u_{nh} and u_{aN} with the conventional method. The virtual Hall signals are corrected before t_5. Then, the load is suddenly increased, and the motor decelerates immediately. Whereas the variation of commutation error cannot be corrected

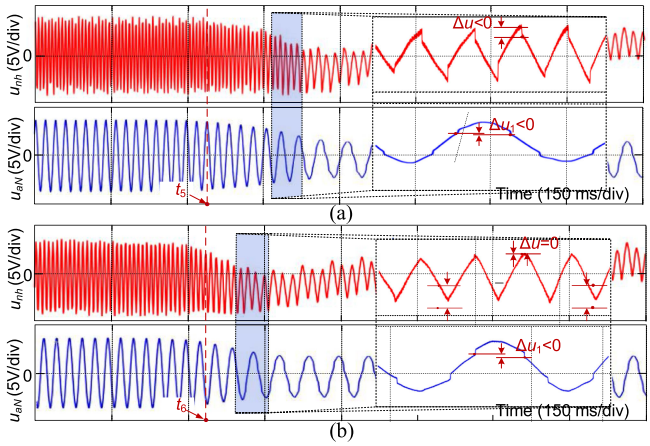


Fig. 19. Comparison of u_{nh} and u_{aN} with different commutation error correction methods when the load increases suddenly at 1200 r/min. (a) Conventional method. (b) Proposed method.

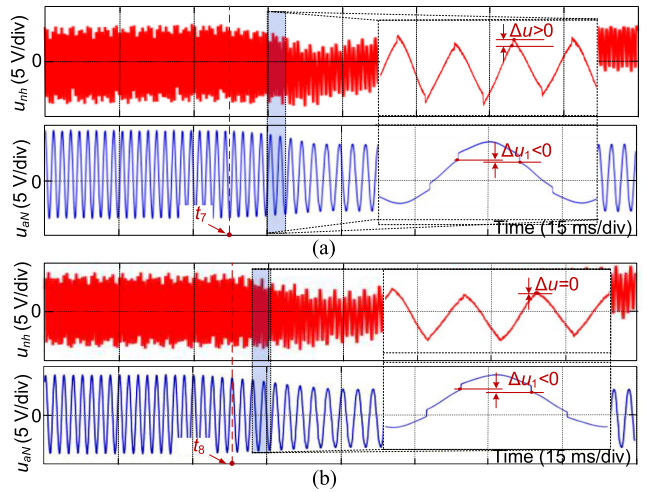


Fig. 20. Comparison of u_{nh} and u_{aN} with different commutation error correction methods when the load increases suddenly at 12 000 r/min. (a) Conventional method. (b) Proposed method.

in time due to the poor dynamic performance. Thus, the corrected angle exceeds the commutation error, which results in the delayed commutation indicated by $\Delta u < 0$. As shown in Fig. 19(b), when the proposed method is adopted, the controlled variable approaches zero, and the commutation error is compensated in time for the significant improvement of the dynamic performance.

From Fig. 6, the rate of the commutation delay angle relative to the speed is small when the motor operates at 12 000 r/min. Thus, the commutation error of the conventional method is mainly generated by the distortion of the controlled variable. Fig. 20(a) shows the waveforms of u_{nh} and u_{aN} with the conventional method, and the commutation is corrected before t_7. With the sudden increase of the load, the motor speed decreases rapidly, which causes Δu_1 to be negative. Thus, the virtual Hall signals are improperly corrected resulting in the advanced commutation error reflected by $\Delta u > 0$. As shown in Fig. 20(b), the commutation is accurate with the proposed method before t_8. During

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the deceleration duration from t_8, Δu remains to be zero, which avoids the improper correction.

Therefore, compared with the conventional method, the proposed method presents a good performance at rapidly varying speed irrespective of the low- or high-speed range.

V. Conclusion

In this article, a fast closed-loop commutation error correction method with anti-interference ability was proposed. First, the commutation errors generated from the conventional method at the rapidly varying speed were analyzed. Then, the proposed method was carried out by constructing a controlled variable first utilized for commutation correction, incorporating feedforward compensation, and using feedback linearization. The experimental results showed that the proposed method cannot only achieve the fast correction of the commutation error at a constant speed, but also present a precise commutation at rapidly varying speed, which improves the motor efficiency.

It is worth noting that the proposed method was very useful in case of rapidly changing speed, such as a sudden increase of load, steep acceleration or deceleration curve. For the BLDC motor with slowly varying speed, the motor signal distortion and variation of the commutation error in the low-speed region can be ignored. Under these circumstances, the conventional method and the proposed method can be both applicable.

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